An AI System for Early Preeclampsia Screening Via Retinal Fundus Photography

A non-invasive, bias averting screening system that detects early retinal biomarkers of preeclampsia with 98% sensitivity

Introduction

1. · Preeclampsia (PE) affects 5%-7% of pregnancies and is the leading cause of maternal morbidity. In the United States, Black women experience a 60% higher risk and more severe outcomes.
2. · Current Screening methods are limited. Traditional screening relies on subjective, race-based risk factors while still failing to identify 40%+ of all positive cases, exacerbating systemic disparities.
3. · Recent research has identified Hypertensive Retinopathy (HR), the visible damage like hemorrhages, and vessel constriction in the retina, as an objective biomarker for PE risk.
4. · Retinal fundus images provide a non-invasive, equitable, and easily accessible diagnostic datapoint in early gestation, often before the onset of severe symptoms typically required for traditional diagnosis.

Engineering Goals

The objective is to create a fully functional, highly accurate, and easily validated screening tool for PE via retinal images...

1. · Model development: Train a Convolutional Neural Network (CNN) model to detect HR with an AUC > 0.90 and Recall > 0.95 for patient safety.
2. · Clinical safety: Implement optimized decision thresholds to minimize false negatives and integrate Image quality constraints to reject low quality (blurry/low-res) images.
3. · Functionality: Create a deployed web app capable of generating results in 8 seconds.
4. · Explainability: Integrate Grad-CAM heatmaps to provide visual evidence of the model's predictions, ensuring clinical trust and accountability.

Acknowledgements

I would like to acknowledge the Kaggle online community for providing public datasets. I would also like to acknowledge Kaggle for providing free NVIDIA Tesla T4 GPUs used to train and test the model. Lastly thanks to Darci Hastings for guidance through the science fair process.

Key Citations

Datasets:
1.Abraham, T. (2018). Diabetic Retinopathy (resized). Kaggle. kaggle.com
2.Cen et al. (2021). Automatic detection of 39 fundus diseases and conditions in retinal photographs using deep neural networks.
3.Nature Communications, 12(1), Article 4828. doi.org
4.Qian et al. (2025). HRDC challenge: a public benchmark for hypertension and HR classification from fundus images. The Visual Computer, 41(2), 1061–1077
5.Washimkar, S. R. (2025). Eye Image Dataset. Kaggle. kaggle.com
6.Maranhão, A. (2020). Ocular disease recognition. Kaggle. kaggle.com

AI Architecture & Transfer

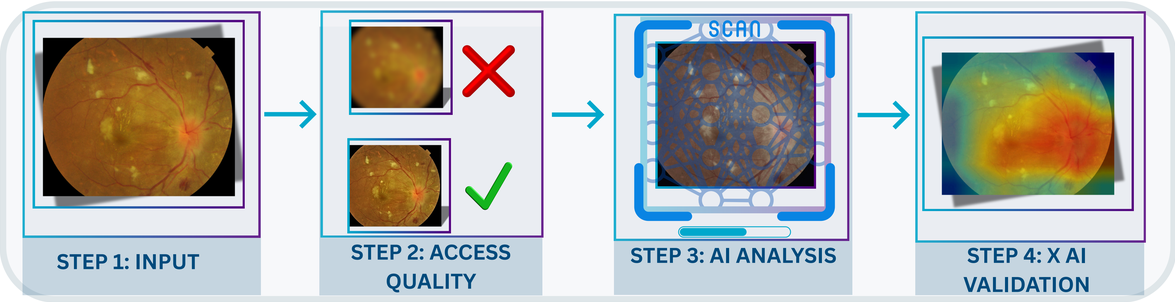
- The study compiled images across six public datasets, including a pretraining set of 35,126 Diabetic Retinopathy images and a training set of 1,020 hypertensive retinopathy (HR) images
- To ensure the ability to generalize various qualities of images; the research utilized Data Augmentation. This technique artificially modifies images by altering hue, brightness, rotation, blurriness and more.
- A Convolutional Neural Network (CNN) is a specialized AI built to analyze visual information. For transfer learning, DenseNet-201 was chosen for its feature-reuse capability, which optimizes pattern recognition in small, complex medical datasets.
- The DenseNet CNN was utilized with a two-step transfer learning approach. Phase 1: trained on Diabetic Retinopathy images to teach the model basic retina features. Phase 2: 1,020 HR images for the final classification task.

System Deployment & Clinical integrity

The final design was integrated into a web application deployed via a cloud-based web interface. This allows researchers to further validate the model and enable clinicians to use it as a screening tool.

Grad-CAM: In medical AI the lack of transparency checks often delays the clinical implementation of AI systems as it erodes trust and accountability within hospitals that utilize AI. That is why this web app integrated explainable AI (XAI) technique called Gradient-weighted Class Activation Mapping (Grad-CAM) which creates visual heatmaps showing what parts of the image the model weighed most significantly for influencing its decision. This allows users to verify that the model is identifying and making decisions on the right areas.

Image constraints: To prevent errors or less accurate diagnoses from suboptimal data, a pre-analysis image quality check was implemented. Using the OpenCV library, the system calculates Laplacian Variance (a mathematical measurement of image sharpness) to detect image blur and monitors the input's resolution. Quality thresholds were based on robustness testing shown in the Performance results section. This checks that the model only processes diagnostic grade images, ensuring consistently high sensitivity despite real-world noisy data.



Graphic created by researcher.

Figure 1: End to end system flow chat with the Image quality gates, AI analysis, and Explainable AI features.

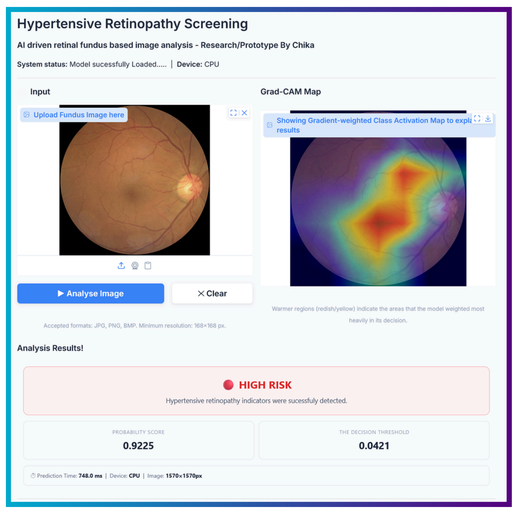


Image taken by researcher.

Figure 2: Screen shot of the functional web application displaying a high-risk prediction and integrated Grad-cam heatmap for visual verification.

Bias mitigation strategy

This project addresses maternal disparities by implementing algorithmic fairness features, eliminating race and pigment-based factors in favor of using the direct, underlying causes of the poorer outcomes.

Pigment-invariant pipeline: This employs standardized color normalization (mean and std scaling) for every input. During data augmentation every batch of images had various color augmentations (brightness, contrast, color jittering) applied. This prevents the model from utilizing retinal pigmentation/color as a factor of analysis, which forces the model to focus strictly on the geometric retinal changes consistent across all populations, regardless of their economic or racial status.

Evidence verification: The integration of Grad-CAM serves as a second preventive measure by allowing clinicians to confirm that the AI's attention is focused on physiological features while helping to support clinicians towards more consistent, visual markings. Overall, these features limit the opportunity for unintentional human error or bias in the diagnosis process.

Results & Clinical Implications

Metric

***AUC: 0.9623**
***Sensitivity: 98.0%**
***Precision: 80.6%**

*AUC is a standard and holistic metric to evaluate AI models. Ranges from 0.0-1.0
*Sensivity measures how many actual positives the model correctly identified. (Traditional methods: 30%-50%)
*Precision measures the likelihood that a predicted positive result is truly positive (Traditional methods: 20%-30%)

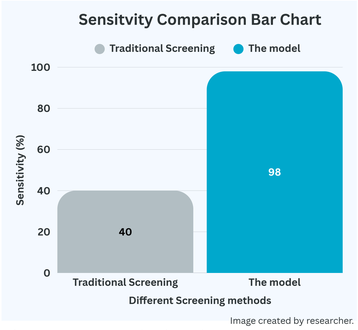


Image created by researcher.

Figure 3: Comparative bar chart analysis showing the designed model (98% sensitivity) significantly outperforming tradition risk-based screenings (30-50%)

Economic Feasibility

Cost effective: The fundus photography equipment for getting the input is relatively cheap ($1k-$5k), already exists in many clinics with nearly zero marginal cost per screening, bypassing the recurring $186 lab fees associated with traditional higher tier models (FMF model). Furthermore, this AI system is completely free and is designed for deployment on standard hardware; ensuring that highly accurate diagnostics are accessible to underfunded rural hospitals.

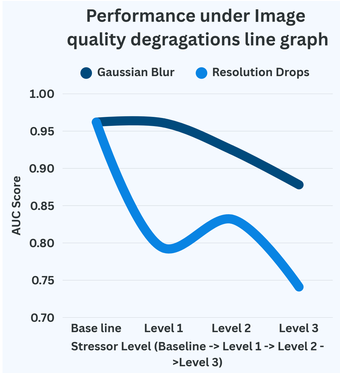


Image created by researcher.

Figure 4: A multi-line graph of the image robustness analysis. While performance remains stable under increased blur, resolution downgrades underscore the necessity of resolution image gates.

Sucessful features

Safety constraints: Successfully set image quality constraints for Resolution (168px) and Blur (39 Laplacian Variance). This ensures each upload is high quality enough for an accurate score.

Efficiency: Mean inference time of 453.52 ms. (0.5 seconds) supports deployment feasibility for high volume or rural clinics. This was powered by a NVIDIA Tesla T4 GPU (16GB GDDR6 VRAM).

Conclusion & Future works

This research demonstrates that a 98% sensitive, Pigment-invariant AI system can significantly outperform reported detection rates of 30-50% in traditional risk-based screening methods. By removing socioeconomic risk factors, this tool provides a scalable, equitable solution to the maternal health crisis. The system is fully deployed with explainable AI heatmaps, image quality checks, and a highly accurate prediction generated in under 500 milliseconds.
As current public datasets lack diverse racial labeling; future work will focus on clinical performance across different racial/ethnic groups who are pregnant. Furthermore, creation and integration with custom-made smartphone-based fundus lenses would be ideal. This could further reduce hardware costs to under $500 and expand access in rural and underserved populations. Lastly, it's important to recognize that this system is designed as a screening aid and not a full diagnostic replacement.